

Abstract

This dissertation explores how video games can be designed to foster empathy and empowerment in the context of forced migration. While existing games often focus on raising awareness, they frequently exclude displaced individuals from the design process. To address this, the project proposes a participatory, AI-assisted storytelling system that allows people with lived migration experience to co-create and replay interactive scenes based on personal memories.

The research follows a three-phase structure: data collection through surveys and a participatory game jam; iterative development of an interview-to-game prototype using a locally run large language model (LLM); and a mixed-methods evaluation. The system includes an interview-based chatbot interface, automatic scene generation, and post-game reflection tools. The evaluation examines the system's emotional, psychological, and representational impact across three player groups: scene authors (migrant participants), other displaced individuals, and players without migration experience.

The project contributes to generative AI research, HCI, and game studies by combining participatory design, storytelling, and technical implementation. It offers both a theoretical framework and a functional prototype to inform future practices in socially responsive game design.

Developing a Video Game for Empathy and Empowerment in the Context of Forced Migration Experiences

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Background & Motivation

With over 110 million people forcibly displaced worldwide due to conflict, climate change, and political instability (UNHCR, 2024), forced migration remains one of the most urgent humanitarian crises of our time. However, the lived experiences of displaced people are often reduced to statistics and policy discussions, which fails to capture the personal, emotional, and psychological dimensions of migration. Mainstream media narratives frequently reinforce stereotypes, portraying migrants either as passive victims or as threats to social stability (Chouliaraki 2017). This signals an urgent need for alternative storytelling approaches that center migrant voices.

Unlike traditional media, video games offer a unique potential for perspective-taking by allowing players to actively experience and shape the narrative. While film or literature presents migration as a fixed story, interactive media can immerse players in decision-making processes and create a sense of agency (Gee 2003). Games such as *Bury Me, My Love* (The Pixel Hunt 2017), which simulates the experience of a Syrian refugee communicating via text messages, or *Papers, Please* (Lukas Pope 2013), which explores the role of immigration bureaucracy, show how game mechanics can highlight complex socio-political realities. Despite the potential of the mentioned games, however, many other migration-focused games maintain an external, observational perspective.

Many empathy-driven games focus primarily on educating outside audiences, leaving little room for agency, co-creation, or self-representation by migrants themselves. The absence of displaced individuals in the game design process results in narratives that, while well-intentioned, may lack authenticity and fail to capture the diversity of migration experiences. I seek to address these gaps by developing a video game that fosters not only empathy but also empowerment in the context of forced migration. The project combines a participatory research method, co-designing interactive narratives with and for displaced individuals, to ensure that they are active contributors rather than passive subjects of their own stories. Furthermore, the work explores the development of a narrative framework with an AI-assisted content generation system that allows players to co-author and replay their own stories.

This dissertation combines insights from HCI, engineering, game studies, psychology, and research in generative AI, to challenge dominant representations of migration and propose an alternative approach based in participatory methodology. It contributes both a theoretical framework for ethical storytelling and a prototype, to demonstrate how storytelling technologies can be designed for social impact.

State of Research & Literature Review

The intersection of video games, forced migration narratives, and participatory design remains underexplored across game studies, psychology, and informatics. While existing research shows that games can foster empathy and perspective-taking (Gee 2003; Smethurst & Craps 2014), most empathy-based games rely on pre-scripted narratives and do not incorporate the voices of those with lived

migration experience. Autobiographical games such as *Bury Me, My Love* (The Pixel Hunt 2017) and *Path Out* (Causa Creations 2017) effectively convey forced migration experiences due to being authored (or co-authored) and developed by people with lived experience. By contrast, many empathy-focused games developed in research contexts are designed about, rather than with, displaced communities.

Participatory design in game studies has gained increasing attention as a means of empowering marginalized communities (Flanagan 2009). Unlike traditional top-down narrative design, participatory methods involve co-creating narratives with affected people. This dissertation builds on existing work by integrating a participatory game jam as a research method (Danilovic et al. 2024). This method aims to bridge the gap between research, community engagement, and game development, allowing people with migration backgrounds to shape their own narratives. Additionally, this study examines the ethical implications of who gets to tell migration stories, which addresses concerns raised in media studies about appropriation and misrepresentation (Ruberg 2020).

AI-assisted storytelling presents new opportunities for personalized game narratives. Although generative AI and co-creative tools are being actively researched at the moment (Liu et al. 2024), little research center player-authored migration stories. Overall, the literature reveals several notable gaps: participatory methodologies are rarely applied in the design of forced migration narratives; most migration-focused games emphasize empathy while overlooking concepts like catharsis and empowerment; and there is limited empirical research on the impact of participatory game design on both players and contributors. This dissertation addresses these gaps by developing a system that combines participatory design, AI-assisted storytelling, and ethical narrative practices, contributing to human-computer interaction, AI, and game studies research.

Research Questions & Objectives

I want to understand how video games can be designed to foster empathy, perspective-taking, and empowerment in the context of forced migration experiences. While previous studies have explored the role of empathy in games, few have examined how individuals with lived migration experiences can actively shape and co-create their own narratives in interactive media. This research seeks to bridge that gap by integrating participatory game design, AI-assisted storytelling systems and mechanics that foster catharsis, into the development of a video game that engages both players with migration backgrounds and those without.

Main research question: How can video games be designed to foster empathy and empowerment in the context of forced migration?

Supporting research questions:

- How can participatory and interactive storytelling systems be used to ethically represent narratives of forced migration?
- How can AI-assisted storytelling systems be designed to meaningfully support the co-creation and replay of personal migration narratives in video games?
- How can the effectiveness of a migration-focused game be measured in fostering empathy and empowering affected communities?

Objectives of the Study

This dissertation contributes to the design, implementation, and evaluation of AI-assisted storytelling systems in games, with a focus on forced migration narratives. Its core objectives are structured around three areas of contribution:

1. Theoretical Contributions

- Develop a participatory design methodology for co-creating narrative games with displaced individuals, grounded in ethical research practices
- Advance the conceptual understanding of catharsis as a game design principle, particularly in the context of displacement and trauma representation.
- Critically analyze dominant representational strategies in commercial games, including how visual and narrative aesthetics can obscure or trivialize serious political themes.

2. Participatory Design

- Organize a participatory game jam¹ as both a research method and a space for co-creating migration narratives with affected individuals. Examine the game jam's effects on empowerment and community-building.
- Investigate the positionality of the researcher through an autoethnographic lens

3. Technical Implementation and Development

- Design and implement an interview-to-game system, using locally run LLMs to transform participant interviews into structured game scenes. Based on this, develop a game prototype that enables interactive storytelling and post-game reflection.
- Evaluate the system's usability and narrative coherence through qualitative and quantitative methods, including player interviews, player behavioral logging, and post-game data analysis.

Methodology and Research Plan

This study employs a mixed-methods research approach, integrating qualitative and quantitative methodologies to explore how video games can foster not only empathy but also empowerment in the context of forced migration. The research follows three phases: data collection, design science (iterative game development), and evaluation of the prototype's impact.

Phase 1: Data Collection

The first phase of this dissertation began with an exploratory qualitative study conducted through a survey with people who have experienced forced migration. This survey focused on participants' perspectives on how their experiences could be represented meaningfully in interactive media. Participants were invited to share personal stories, suggest narrative themes, and reflect on what kinds of gameplay elements they would want to see in a game based on their lived experience. Importantly, all participants who shared their stories expressed a desire for those narratives to be included in the final game. These responses informed the development of a game design framework, which in turn guided the creation of an interview-to-game system that transforms participant interviews into interactive scenes (Harshina et al. 2024). However, the limited diversity of participant backgrounds prompted us to seek an additional data

¹ A collaborative, time-limited event where participants rapidly prototype games, often around a shared theme.

collection approach. To address this, a more embedded, community-focused participatory method was introduced: the game jam. The participatory game jam is designed to serve both as a research tool and as a community-building initiative. It invites people with and without forced migration experiences to co-create game narratives. The ethical design of the game jam includes attention to informed consent, participant safety, and recognition of narrative ownership. Its outputs will inform both the game prototype and the storytelling system developed in Phase 2.

Additionally, challenges encountered during this early phase, including the personal proximity of the researcher to the topic, led to the formation of a collaborative autoethnographic study with a researcher from the University of Bremen. This ongoing collaboration with other researcher(s) working on similarly personal and ethically complex projects aims to create guidelines for researchers who are personally involved in the communities they study.

Phase 2: Iterative Development of the Prototype

Building on the insights from Phase 1, the second phase focuses on the iterative development of a game prototype designed to transform interviews into interactive scenes. At the core of this phase is the development of an interview-to-game system designed to support co-creation by transforming participant stories into interactive scenes using AI-assisted storytelling. The system consists of the following components:

1. Interview Interface – an AI-assisted chatbot (developed in *Gradio*ⁱ) guides participants through an interview focused on memory, identity, and experience. A locally running large language model (e.g., *Zephyr*ⁱⁱ) parses the interview and identifies key narrative elements, such as character roles, emotional tone, and environmental settings.
2. Game Scene Generation – players engage with game scenes through a roleplaying chatbot interface. Migrant participants explore experiences with restored agency; others interact with scenes under restricted conditions to emphasize systemic limitations. Visual prompts are generated using *Freepik*'s ethical image APIⁱⁱⁱ. The system is built for future integration with narrative engines like *Ren'Py*^{iv}.
3. Post-Game Reflection – after each session, players (both authors and non-authors) are invited to write reflections. These are used in Phase 3 to evaluate emotional resonance, identification, and narrative impact.

The development will follow design science principles and a heuristic evaluation cycle, incorporating feedback from researchers, designers, and play-testers. Outputs from this phase are evaluated in Phase 3 through a mixed-methods study.

Phase 3: Evaluation Study on the Effects of the Prototype

Once the prototype reaches a refined stage, the research moves into an evaluation phase, where user studies, playtesting, and qualitative research methods will be used to analyze how players engage with migration narratives in the game. This phase will explore whether players' perceptions of migration change after gameplay and examine the impact of different narratives and game mechanics on fostering empathy, empowerment, and critical reflection.

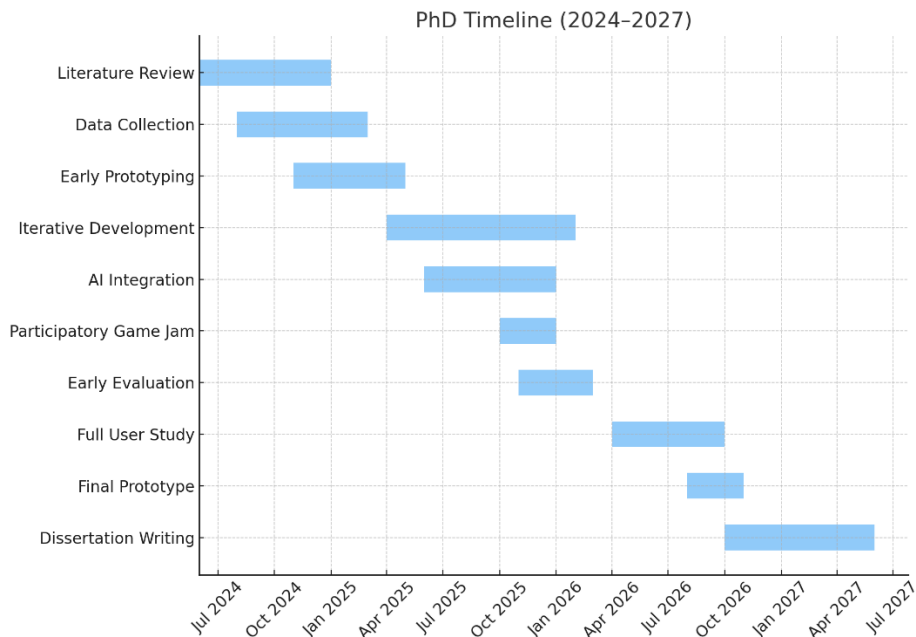
The evaluation is designed to analyze the effect of the game across three distinct player groups:

1. Scene Authors (Migrant Participants) – How does engaging with their own authored scene impact the original storyteller? This includes potential for catharsis, restored agency, and narrative empowerment.
2. Other Players with Forced Migration Experience – How do participants respond to scenes authored by others? Does the game support emotional resonance, recognition, or solidarity across shared experiences?
3. Non-Migrant Players – How does constrained gameplay affect empathy, perspective-taking, and understanding of systemic limitations?

The study uses the following evaluation methods:

- Player Behavior Tracking – the game will log player choices and interactions with characters to assess how different mechanics influence engagement and perspective-taking.
- Post-Game Reflection Logging – players are encouraged to write or save their reflections after each memory scene. These written reflections will be analyzed qualitatively to assess emotional resonance and personal meaning-making.
- Qualitative Interviews with Migrant Participants – people with lived experience of forced migration will be interviewed about their gameplay experience, focusing on narrative authenticity, emotional impact, and their sense of agency or catharsis during interaction. These interviews aim to validate whether the system supports ethical self-representation

Timeline and Publications



Theoretical Contributions

Venue & Year	Title / Focus	Status
Future Realities of Gaming 2024 (Full Paper) + Games and Culture (Journal)	<i>Unraveling the Normalization of Colonial, Imperial and Authoritarian Narratives in Modern Video Games</i>	Accepted, Planned for 2026

Venue & Year	Title / Focus	Status
International Conf. on Games and Narrative 2025 (Extended Abstract)	<i>For Empowerment's Sake: Rethinking Video Game Adaptations of Real-Life Experiences</i>	Published 2025
CHI 2026 (Full Paper)	<i>Empowerment and Catharsis in Game Design</i>	Under revision, planned for 2026

Methodological Contributions

Venue & Year	Title / Focus	Status
GALA 2024 (Short Paper)	<i>Experiencing Forced Migration: A Participatory Approach to developing a Serious Video Game</i>	Published 2024
CHI PLAY 2024 (Extended Abstract)	<i>Developing a Virtual-Reality Game for Empathy Enhancement in the Context of Forced Migration Experiences</i>	Published 2024
CHI 2026 (Full Paper)	<i>Autoethnographic study</i>	Study ongoing, planned for 2026
CHI PLAY 2026 (Full Paper)	<i>Participatory Methodologies from the Game Jam</i>	Planned for 2026-2027

Technical Contributions

Venue & Year	Title / Focus	Status
Experimental AI in Games Workshop 2024 (Short Paper)	<i>A Generative AI Game as a Tool for Research in AI Creativity and Sustainable Design</i>	Published 2024
IEEE Conference on Games (Full Paper)	<i>Technical implementation and evaluation of the interview-to-game system</i>	Planned for 2026-2027

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ⁱ Gradio is an open-source Python library: <https://www.gradio.app/>

ⁱⁱ Zephyr is a large language model: <https://huggingface.co/HuggingFaceH4/zephyr-7b-alpha>

ⁱⁱⁱ Freepik provides a developer API for access to their ethical image database: <https://docs.freepik.com/introduction>

^{iv} Ren'Py is an open-source video game engine: <https://www.renpy.org/>